Original Article

Design, Synthesis, and Biological Evaluation of Bis-Imine-Based Metal Complexes

Nirmal Joshi[1] | Saroj Bembalkar[1,*] | Smita Rajani[2] | Rajendra Pawar[3] | Jagannath Godse[4]

[1]*Department of Chemistry, Deogiri College, Aurangabad 431005, Maharashtra, India*
[2]*Shree Ramkrishna Institute of Computer Education and Applied Sciences, Surat 395001, Gujarat, India*
[3]*Department of Chemistry, Shiv Chhatrapati College, Aurangabad 431003,Maharashtra, India*
[4]*Department of Engineering Science, Hi-Tech Institute of Technology, MIDC Waluj, Aurangabad 431136 Maharashtra, India*

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ABSTRACT

The new bis-imine ligand was synthesized by the condensation reaction of 1-((3-(4-fluorophenyl)-1-isopropyl-1H-indol-2-yl)methylene) hydrazine with ortho-vanillin. The synthesized ligand was subsequently employed to produce diverse transition metal complexes through the combination with suitable metal salt solutions on subsequent heating. Using UV-Vis, FTIR, ^{1}H-NMR spectroscopy, and mass spectrometry, the functional groups, coordination interaction, and probable morphologies of the synthesized compounds were examined. On spectral data basis, it was confirmed that the ligand and its metal complexes were made. The ligand and its metal complexes were evaluated to determine their potential biological consequences. The biological activity results showed that specific metal complexes were more effective than the free ligand. Certain combinations exhibited promising effectiveness against antibacterial, anti-tuberculosis, and antimalarial activities. The results indicate that coordination of the bis-imine ligand with metal ions improves its suitability for biological activity. The indole-based bis-imine metal complexes could be ideal candidates for developing new biologically active molecules for medical applications.

Introduction

In the biological system, metal complexes of bis-imine ligands are important [1-3]. Because the substrate binds tobiologically active site of enzyme. Bis-imines or bis Schiff bases are compounds that contains two azomethine (–CH= N–) group and are used in medicinal and chemical fields. Bis-imine metal complexes exhibit biological activities, including antibacterial, antituberculosis, anticancer, and antifungal activities [4]. Bis-imines [5] can be used for manufacture of important medication [6,7]. As these ligands are bi or tri dentate, they form many transition metal complexes [8,9]. Bis-imine metal complexes are nowadays employed for chemical reagents and also to cure some diseases [10,11]. Bis-Schiff base complexes with metal atoms fascinated the thoughts of chemistry researchers due to the applications of

*Corresponding Author: s_bembalkar@yahoo.com (S. Bembalkar)

these bis-imines as anti-inflammatory, antimicrobial, antiviral activities, and so on. Bis imines and complexes with metal containing halogens [12,13] showed antimicrobial activity. Mercury and cadmium complexes with 2, 6- bis (benzinidazol-2-yl) pyridine ligand display antimicrobial effect. Bis-imine metal complexes are used as catalyst such as hafnium and zirconium imino-amido complexes used as catalyst in polymerisation of ethylene [14-16].

The aim of this research work was to prepare a new bis-imine ligand. The synthesis of this ligand was achieved through the condensation reaction of (1-((3-(4-fluorophenyl) -1-isopropyl -1*H*-indol-2-yl) methylene) hydrazine) with ortho vanillin. Spectroscopic techniques were used to characterized the synthesised ligand. Complexes of metal of Zn, Cu,, Ni, Co, Hg, Mn, Sn, Fe, Cd, and Zr are prepared from ligand and tested for their ability to fight against microbial, tuberculosis, and malaria.

Materials and Methods

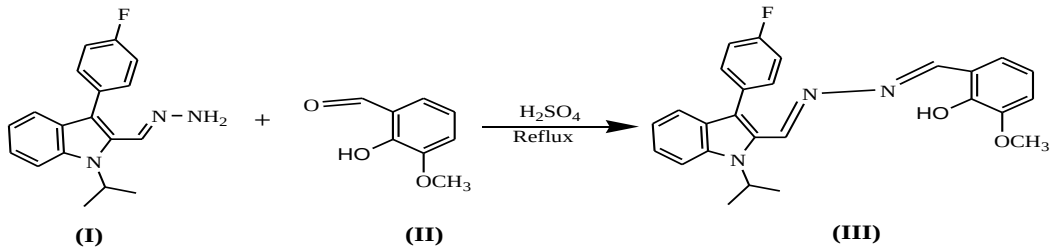
All the purchased chemicals and solvents were of analytical grade and used without further purification.

Synthesis of Bis-Imine Ligand

An equimolar mixture of (*E*)-1-((3-(4-fluoro-phenyl)-1-isopropyl-1*H*-indol-2-yl) methylene) hydrazine (I) (1mmol, 0.295 g) and Ortho Vanillin (II) (1mmol, 0.152 g) was mixed in solvent of ethyl alcohol and 1 or 2 drops concentrated sulfuric acid was added as catalyst to the reaction mixture. The resulting mixture was refluxed for 4 - 5 hs. On completion of process, the reaction mixture was cooled. The obtained product bis-imine ligand (III) was dried after filtration and recrystallized in ethyl alcohol (Scheme 1).

FTIR Spectroscopy Study of Bis-Imine Ligand

The FTIR spectroscopy of bis-imine ligand is displayed in Figure 1 and Table 1 reports the infrared spectroscopy data of ligand.



Scheme 1: Synthesis of bis-imine ligand

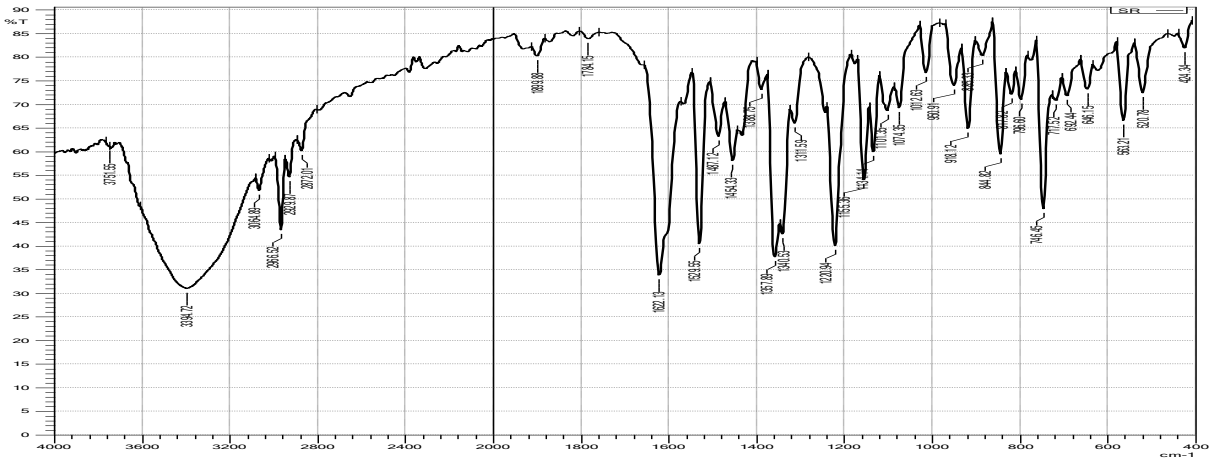


Figure 1: The FTIR spectroscopy of bis-imine ligand

Table 1: FTIR spectroscopy data of bis-imine ligand

Compound	ν cm^{-1} (Ar—H)	ν cm^{-1} (C=N)	ν cm^{-1} (C=C)
Ligand	2,966	1,622	1,529

The FTIR spectrum of the synthesised bis-imine ligand exhibited that the Schiff base framework was successfully formed by showing unique vibrational bands. The azomethine (C=N) stretching vibration was a strong sign that the imine bond was formed during the reaction. It appeard as a strong absorption band in the range of 1,622 cm^{-1}. There were also bands in the 1,529 cm^{-1} range that showed aromatic C=C stretching vibrations. This indicates that the ligand structure had a ring of phenyl. The absorptions at 2,966 cm^{-1} were caused by C–H stretching vibrations in aromatic and aliphatic compounds, respectively.

Mass Spectrometry Study of Bis-Imine Ligand

The molecular mass of compound was determined by mass spectrometry. The calculated exact mass of bis-imine ligand was 429 and the molecular ion peak (M+1) was 430. The mass spectrum of ligand is depicted in Figure 2.

^{1}H-NMR Study of Bis-Imine Ligands

The ^{1}H-NMR spectroscopy bis-imine ligand is demnstrated in Figure 3 and Table 2 reports the ^{1}H-NMR data of the ligand.

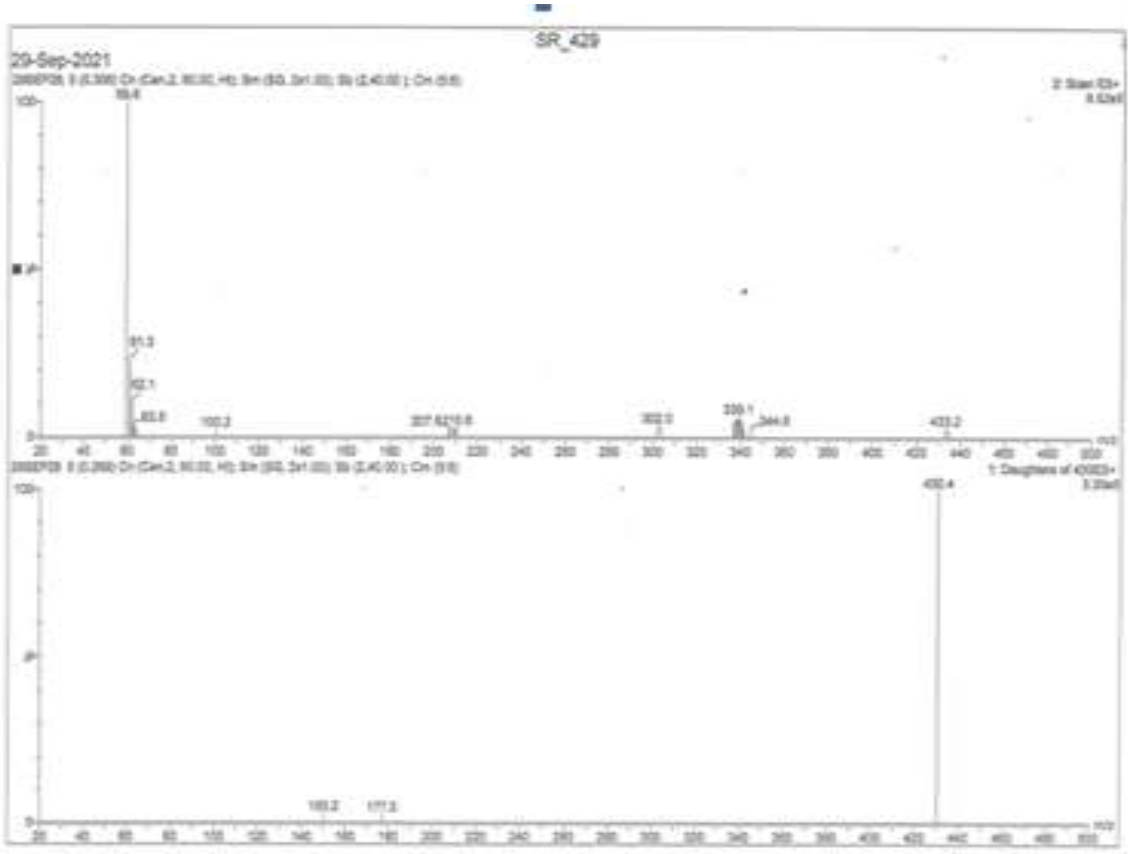


Figure 2: Mass spectrum of bis-imine ligand

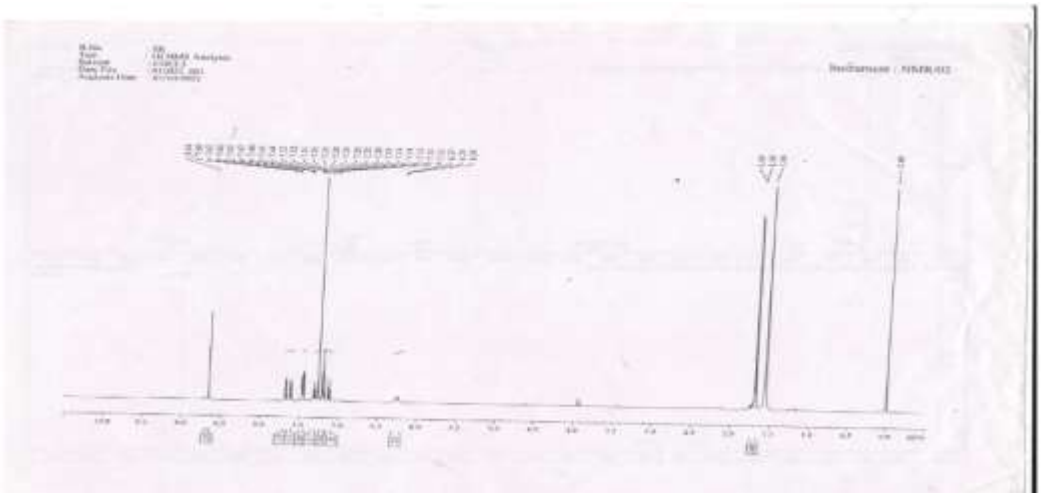


Figure 3: ^{1}H-NMR spectroscopy of bis-imine ligand

Table 2: ^{1}H-NMR spectroscopy data of bis-imine ligand

Assignment for ligand (L_4)	HC=N	C-H(Ar)	-OMe	OH
Chemical shift (δppm)	8.66	7.11-7.68	3.89	4.5

^{1}H-NMR spectroscopy verified the formation of the bis-imine ligand. The azomethine proton (–CH=N–) showed a typical singlet in the range of δ 8.66 ppm. This proves that the aldehyde and diamine precursors have come together. The aromatic protons in the phenyl rings appeared as a series of multiplets between δ 7.11 and 7.68 ppm. This is due to the ligand's substituted aromatic framework with different proton environments. The signals 3.89 ppm came from the methylene protons (–CH_2–) that were next to nitrogen atoms in the diamine backbone. There were also resonances in the δ 1.0–1.8 ppm range that match aliphatic methyl or methylene protons that were part of the ligand structure. The absence of a characteristic aldehydic proton signal in the δ 9–10 ppm range further indicated the completion of the Schiff base condensation reaction. Small signals near δ 2.50 ppm and δ ~3.30 ppm were caused by leftover solvent (DMSO-d_6) and small amounts of water. The spectral features generally aligned with the proposed structure of the synthesized bis-imine ligand, indicating the successful formation of the azomethine functionality.

Synthesis of Metal Complexes

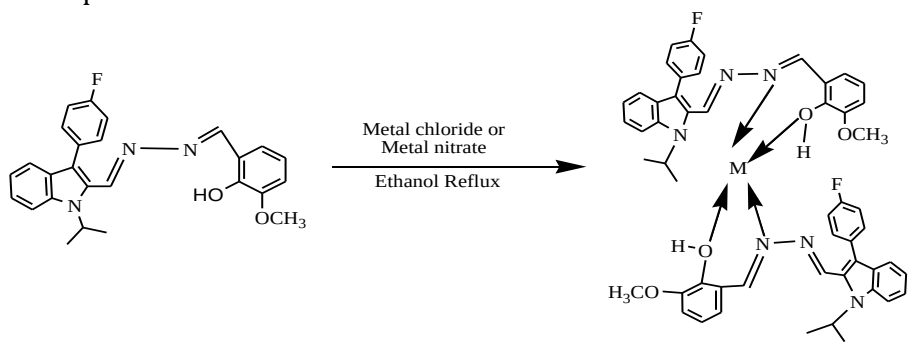
A solution of metal salt in ethyl alcohol (nickel nitrate, copper nitrate, cobalt nitrate, zinc chloride, and manganese chloride) was mixed with a hot ethanolic solution of bis-imine ligand. The ratio of ligand to metal salt in mmol was 2:1. To the mixture solution, ammonia was added to maintain pH of reaction mixture alkaline (pH = 7–8). The mixture was further heated for 7 hours. After the reaction was over, ice was added to chill the product and the precipitation was filter, rinsed using ice water, and then allowed to dry (Scheme 2). The synthesized ligand and its metal complexes physical and elemental analytical data are shown in Table 3.

Results and Discussion

FTIR Spectral Study of Bis-Imine Ligand Metal Complexes

For C—H bonds, the stretching frequency bands in the IR spectrum of bis-imine metal complexes were found between 2,962 and 2,996 cm^{-1}. The bands between 565 and 796 cm^{-1} were the M-O stretching frequency, which was when the Ligand's oxygen and the metal (*i.e.,* for metal ions and ligand) were connected. The bands that showed up between 1,602 and 1,627 cm^{-1} were for the frequency of C=N group. Frequency bands between 420 and 500 cm^{-1} showed M—N bonds, which meant that the nitrogen atom for ligand and metal atom were connected.

Significant IR spectral bands and their evaluations are provided in Table 4.



Ligand

M= Cu(II), Ni(II), Co(II), Cd(II), Zr(II), Fe(II), Zn(II), Mn(II), Hg(II), Sn(II).

Scheme 2: Synthesis of metal complexes

Table 3: Physical and analytical data of the synthesized compounds

Sr./ No.	Compound	Mole. formula	Mole. weight	M.P. (˚C)	Colour	Elemental analysis % calculated			
						C %	H%	N %	M %
1	Bis-imine ligand (L_4)	$C_{26}H_{24}N_3O_2F$	429.49	210	Yellow	72.6	5.58	9.77	-
2	Zn Complex	$C_{52}H_{48}F_2N_6O_4Zn$	924.36	300	Yellow	67.57	5.23	9.09	7.07
3	Cu Complex	$C_{52}H_{48}F_2N_6O_4Cu$	922.52	>300	Yellowish Brown	67.70	5.24	9.11	6.89
4	Ni Complex	$C_{52}H_{48}F_2N_6O_4Ni$	917.67	280	Yellow	68.06	5.27	9.16	6.40
5	Co Complex	$C_{52}H_{48}F_2N_6O_4Co$	917.91	>300	Brown	68.04	5.27	9.16	6.42
6	Mn Complex	$C_{52}H_{48}F_2N_6O_4Mn$	913.91	>300	Yellowish Brown	68.34	5.29	9.20	6.01
7	Hg Complex	$C_{52}H_{48}F_2N_6O_4Hg$	1059.56	>300	Lemon Yellow	58.94	4.57	7.93	18.93
8	Cd Complex	$C_{52}H_{48}F_2N_6O_4Cd$	971.38	>300	Lemon Yellow	64.30	4.98	8.65	11.57
9	Sn Complex	$C_{52}H_{48}F_2N_6O_4Sn$	977.68	270	Lemon Yellow	63.88	4.95	8.60	12.14
10	Zr Complex	$C_{52}H_{48}F_2N_6O_4Zr$	950.2	>300	Lemon Yellow	65.73	5.09	8.84	9.60
11	Fe Complex	$C_{52}H_{48}F_2N_6O_4Fe$	914.82	>300	Ivory	68.27	5.29	9.19	6.10

Table 4: Infra-red spectra data of ligand and metal complexes

Compound	ν cm^{-1} (C=N)	ν cm^{-1} (C—H)	ν cm^{-1} (M—O)	ν cm^{-1} (M—N)
Ligand Bis-Imine (L_4)	1,622	2,966	-	-
Zn Complex	1,604	2,972	565	453
Cu Complex	1,604	2,962	565	470
Ni Complex	1,604	2,970	667	500
Co Complex	1,604	2,972	694	500
Mn Complex	1,604	2,978	750	430
Hg Complex	1,624	2,962	750	500
Cd Complex	1,616	2,964	796	430
Sn Complex	1,618	2,976	657	466

Zr Complex	1,627	2,966	750	420
Fe Complex	1,602	2,964	748	455

UV Spectral Analysis of Metal Complexes

The λ_{max} reading of the UV-Vis spectrum of synthesized metal compounds is presented in Table 5. In DMSO solvent, ultraviolet visible spectrum [17] for these compounds was recorded.

Table 5: λ_{max} value of the synthesized metal complexes

Compound	Wavelength (λ_{max})
Zn Complex	231.80
Cu Complex	614.20
Ni Complex	608.60
Co Complex	475.40
Mn Complex	496.50
Hg Complex	513.00
Cd Complex	498.40
Sn Complex	501.00
Zr Complex	498.00
Fe Complex	405.60

Biological Activity

Antimicrobial Study

Each metal complexes was tested against three types of fungi (*A. niger* MTCC 282, *A. clavatus* MTCC 1323, and *C. albicans* MTCC 227) and four types of bacteria (*S. pyogenus* MTCC 442, *S. aureus* MTCC 96, *E. coli* MTCC 443, and *P. aeruginosa* MTCC 1688). Mueller–Hinton agar was utilized as the nutrition media. The Broth dilution method was employed to determine antibacterial activity. The solutions of synthesized metal complexes was prepared in DMSO solvent. Broth Hinton method was employed for cultivating microorganisms, also to thin out the compound solution of the microorganisms for testing. The sample tubes were kept at 37 °C overnight. DMSO served as the control. A control tube that did not have a test substance, was sub-cultured with media to let microorganisms grow and was then incubated overnight to 37 °C. MIC, or Minimum inhibitory concentration, of control test microorganism was noted to examine the correctness of the metal complexes' concentrations. The MIC was noted as the lowest concentration that stopped the development of microorganisms. Afterwards, the MIC of newly made metal complexes [18–20] was compared to the MIC of typical medications for bacteria and fungus (*i.e.,* ampicilin) (Table 6).

Antituberculosis Activity

The cultures of *Mycobacterium tuberculosis* (H37Rv strain) were examined against all of the novel metal complexes. The minimal inhibition concentration was used to determine how well the drug worked against tuberculosis. These tests for bacterial susceptibility were carried out in a bottle. L.J. nutrition medium was utilized to cultivate the microorganism. To acquire the right concentration of test chemicals, DMSO solvent was used. For primary and secondary screening, in primary as well as secondary testing, a series of dilutions was made [21-23]. Inoculum size were set at 1 mg/mL for the test strain. The minimal inhibition concentration (*i.e.,* MIC) was the least concentration that caused a least 99% inhibition. The MICs of the synthesised compounds were recorded and compared to Isoniazid and Rifampicin, which are conventional medications, as indicated in Table 7.

Table 6: Antimicrobial activity of the synthesized metal complexes

Compounds	Antibacterial activity MIC				Antifungal activity MIC		
	S. pyogenus	*S. aureus*	*E. coli*	*P. aeruginosa*	*A. niger*	*A. clavatus*	*C. albicans*
ZnL_2	62.5	100	125	100	1,000	1,000	500
CuL_2	125	125	62.5	50	500	1,000	500
NiL_2	500	500	100	125	500	1,000	250
CoL_2	125	100	100	125	1,000	1,000	500
Mn L_2	100	50	12.5	25	100	500	250
Hg L_2	100	100	125	250	1,000	1,000	500
CdL_2	100	25	50	100	1,000	500	1,000
SnL_2	100	62.5	100	50	1,000	1,000	1,000
ZrL_2	500	250	250	500	500	500	500
FeL_2	500	125	125	250	>1,000	>1,000	1,000
Ampicillin	100	250	100	-	-	-	-
Chloramphenicol	50	50	50	50	-	-	-
Nystatin	-	-	-	-	100	100	100
Greseofulvin	-	-	-	-	100	100	500

Table 7: Anti-tuberculosis activity

Bactria	*$H_{37}RV$*
Compound	MIC μg/mL
ZnL_2	250
CuL_2	125
NiL_2	250
CoL_2	100
MnL_2	125
HgL_2	100
CdL_2	500
SnL_2	100
ZrL_2	100
FeL_2	125
Isoniazid	0.20 μg/mL 99% inhibition
Rifampicin	40 μg/mL 99% inhibition

Antimalarial Activity

This *in vitro* experiment for assessing the vulnerability of *P. falciparum* for antimalarial metal compounds was conducted in 96-well microtitre plates. The antimalarial activity was examined in synthesized metal complexes. The *P. falciparum* strain was cultivated using RPMI 1640 culture media [24,25]. DMSO was used to dilute the test substances. Table 8 presents the antimalarial activity of synthesised metal complexes. The MIC was the lowest amount of test substance that stopped the microorganism from growing. Standard medications included chloroquine and quinine.

Table 8: Anti-malarial activity

Compound	Mean IC_{50} values
ZnL_2	1.41 μg/mL
CuL_2	1.10 μg/mL
NiL_2	1.62 μg/mL
CoL_2	0.38 μg/mL
Mn L_2	0.97 μg/mL
Hg Complex	0.65 μg/mL
Cd Complex	0.93 μg/mL
Sn Complex	1.28 μg/mL
Zr Complex	0.88 μg/mL
Fe Complex	0.96 μg/mL
Chloroquine	IC_{50}-0.020 μg/mL
Quinine	IC_{50}-0.268 μg/mL

A new bis-imine ligand was prepared by condensing 1-((3-(4 -fluorophenyl) -1-isopropyl-1H-indol -2-yl) methylene) hydrazine with *Ortho* Vanilin. Bis imine is further utilized for the preparation of metal complexes. Spectroscopic approaches are used to confirm newly prepared compounds, such as ligands and all metal complexes.

The IR spectra of the ligand display a band at 1,622 cm^{-1}, corresponding to the stretching frequency of the C=N group. The spectral band between 420 and 500 cm^{-1} shows the M-N bands frequency for the ligand's metal and nitrogen. The IR frequency spectrum from 2,962 to 2,996 cm^{-1} is the same as the frequency, at which C-H bonds stretch. The M—O stretching frequency can be observed in the bands between 565 and 796 cm^{-1}. The ligand acts as bidentate because it connects to metal through one nitrogen atom and one oxygen atom in its structure. The λ_{max} value of metal complexes in UV spectra is between 231 nm and 614 nm. The λ_{max} reading for copper complex was higher when it was absorbed. Complexes of metal *i.e.,* zinc, nickel, cobalt, cadmium, mercury, manganese, tin, iron, zirconium, and copper have metal: ligand 1:2 ratios.

Metal complexes are tested against microbial, tuberculosis, and malaria. When metal complex was tested to its activity to kill bacteria, it was found that manganese and zinc are quite effective against MTCC 442 *i.e., S. pyogenus*, *S. aureus* MTCC 96 was affected by cobalt, manganese, and zinc. Copper cobalt and nickel worked very well opposed to *E. coli* MTCC 443, while Zn(II), Cu(II), and Mn(II) worked very well against *P. aeruginosa* MTCC 1688, as compared to normal medications. Cu(II) worked against *A. niger* MTCC 282. Mn(II) showed action against *A. clavatus* MTCC 1323. Zn(II), Cu(II), Co(II), and Mn(II) exhibited moderate to superior efficacy against *C. albicans* MTCC 227 compared to traditional pharmaceuticals. The prepared complexes of Metal Hg, Cu, Zr, Sn, Mn, Co, and Fe exhibited superior antituberculosis action against the *H37Rv* strain of *Mycobacterium tuberculosis* when compared to conventional medicines, such as isoniazid and rifampicin. Co(II), Hg(II), Cd(II), Zr(II), Mn(II), and Fe(II) metal complexes were far better at fighting malaria than traditional medications including quinine and chloroquine.

Conclusion

In this study, a new bis-imine Schiff base ligand was successfully synthesised via the condensation reaction between (*E*)-1-((3-(4-fluorophenyl)-1-isopropyl-1*H*-indol-2-yl)methylene) hydrazine and ortho-vanillin in an ethanolic medium, utilizing concentrated sulfuric acid as a catalyst. The synthesized ligand was subsequently utilized to prepare a series of transition metal complexes with Zn(II), Cu(II), Ni(II), Co(II), Mn(II), Hg(II), Cd(II), Sn(II), Zr(II), and Fe(II) metal ions in a 1:2 metal-to-ligand

ratio. Biological testing showed that some of the metal complexes showed strong antimicrobial effects against bacterial strains such as *S. pyogenus, S. aureus*, *E. coli*, and *P. aeruginosa*, as well as fungal strains such as *A. niger*, *A. clavatus*, and *C. albicans*. Notably, Mn(II), Zn(II), and Cu(II) complexes exhibited significant antibacterial properties. The complexes also showed antituberculosis activity against the H37Rv strain of *Mycobacterium tuberculosis*, with some complexes showing better activity than standard drugs. The synthesised complexes also had moderate to strong antimalarial activity against *Plasmodium falciparum*. The Co(II), Hg(II), Cd(II), Zr(II), Mn(II), and Fe(II) complexes were better at stopping the parasite from growing. These results indicate that Schiff base metal complexes could be viable candidates for the formulation of innovative antimicrobial, antitubercular, and antimalarial agents.

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ORCID

Nirmal Joshi
https://orcid.org/0009-0001-7198-7928
Saroj Bembalkar
https://orcid.org/0000-0001-8249-6246
Smita Rajani
https://orcid.org/0009-0005-6040-1425
Rajendra Pawar
https://orcid.org/0000-0002-5861-2080
Jagannath Godse
https://orcid.org/0000-0002-3227-3885

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